1. Linear Algebra & Calculus

Vectors & Matrices

- **Inner Product:** $\theta^\top x = \sum_{k=1}^{d} \theta^{(k)} x^{(k)}$ (Scalar). **Transpose:** $(AB)^\top = B^\top A^\top$. **Symmetric Matrix:** $A^\top = A$. **Euclidean Norm Squared:** $\|v\|_2^2 = v^\top v$. **Cauchy-Schwarz Inequality:** $(\sum x_i^2)(\sum y_i^2) \geq (\sum x_i y_i)^2$.
- **Quadratic Exp:** $(x+y)^\top A(x+y) = x^\top Ax + 2x^\top Ay + y^\top Ay$ (if A is symmetric).

Matrix Calculus Rules

- For a vector θ and symmetric matrix A: **Constant:** $\nabla_\theta(c) = 0$. **Linear:** $\nabla_\theta(b^\top\theta) = b$. **Quadratic:** $\nabla_\theta(\theta^\top A\theta) = 2A\theta$.

2. Optimization & Convexity

Convexity Definitions

- **Convex Function:** The line connecting two points on the curve lies above the curve. **Sums:** The sum of convex functions is convex. **1D test:** Second derivative $f''(\theta) \geq 0$. **Multi-D test:** Hessian matrix is **Positive Semi-Definite (PSD)**. (Eigenvalues ≥ 0).

First-Order Optimality Condition

- For an unconstrained convex optimization problem, the global minimum is found where the gradient is zero: $\nabla\ell(\hat{\theta}) = 0$.

3. Univariate Regression (1D)

Without Intercept & With Intercept

- Given N pairs of $(\hat{x}_i, \hat{y}_i)$. **Case 1: Without Intercept: Model:** $\hat{y} = \theta\hat{x}_i$. **Loss Function (Sum of Sq. Errors):** $\ell(\theta) = \sum_{i=1}^{N} (\theta\hat{x}_i - \hat{y}_i)^2$. **Optimal Slope:** $\hat{\theta} = \frac{\sum_{i=1}^{N} \hat{x}_i\hat{y}_i}{\sum_{i=1}^{N} \hat{x}_i^2}$.
- **Case 2: With Intercept: Model:** $\hat{y} = \theta_0 + \theta\hat{x}_i$. **Loss Function:** $\ell(\theta_0, \theta) = \sum_{i=1}^{N} (\theta_0 + \theta\hat{x}_i - \hat{y}_i)^2$. **Averages & Sample Cov/Var:** $\bar{x} = \frac{1}{N}\sum \hat{x}_i$, $\bar{y} = \frac{1}{N}\sum \hat{y}_i$, $S_{xx} = \frac{1}{N}\sum(\hat{x}_i - \bar{x})^2$, $S_{xy} = \frac{1}{N}\sum(\hat{x}_i - \bar{x})(\hat{y}_i - \bar{y})$. **Optimal Parameters:** $\hat{\theta} = \frac{S_{xy}}{S_{xx}} = \frac{\sum(\hat{x}_i - \bar{x})(\hat{y}_i - \bar{y})}{\sum(\hat{x}_i - \bar{x})^2}$, $\hat{\theta}_0 = \bar{y} - \hat{\theta}\bar{x}$.
- **2-Point Fit Trick:** If given only 2 points, Slope $\hat{\theta} = \frac{\Delta y}{\Delta x}$, Intercept $\theta_0 = y - \hat{\theta}x$ (plug in either point).

4. Multivariate Regression

Vectorized Form & Solutions

- For d-dimensional features, we vectorize the data. $X \in \mathbb{R}^{N\times d}$ (Data Matrix), $Y \in \mathbb{R}^{N\times 1}$ (Response Vector), $\theta \in \mathbb{R}^{d\times 1}$ (Parameter Vector).
- **Loss Function Expansion:** $\ell(\theta) = \|X\theta - Y\|_2^2 = (X\theta - Y)^\top(X\theta - Y) = \theta^\top X^\top X\theta - 2Y^\top X\theta + Y^\top Y$.
- **Gradient:** $\nabla\ell(\theta) = 2X^\top X\theta - 2X^\top Y$.
- **First Order Condition:** $X^\top X\hat{\theta} = X^\top Y$.
- **Analytical Solution (Normal Equation):** $\hat{\theta} = (X^\top X)^{-1}X^\top Y$.
- **Augmentation Trick (To add intercept):** Add a column of 1's to X (making it $N \times (d+1)$), and prepend θ_0 to θ. The Normal Equation remains exactly the same.

5. Key Regression Properties

Core Concepts

- **Invertibility & Small-Sample Problem:** $\hat{\theta}$ requires inverting the **Gram Matrix** $X^\top X$. $X^\top X$ is invertible *iff* X has **full column rank**. **Small-Sample Problem:** If $N < d$ (fewer observations than features), $X^\top X$ is NOT invertible. The problem is ill-posed and has an infinite number of optimal solutions.
- **Orthogonality Condition:** Let $E = Y - X\hat{\theta}$ be the vector of errors. $X^\top E = 0$. The vector of errors is orthogonal to the column span of the feature matrix X.
- **Maximum Likelihood Estimation (MLE):** If we assume errors are independent and follow a Normal Distribution $\mathcal{N}(0, \sigma^2)$: Maximizing the probability (Likelihood) of the data is mathematically equivalent to minimizing the **sum of squared errors**. **Takeaway:** OLS estimator = MLE estimator under Gaussian noise.

6. Homework 1 Variants

Transformations & Comparisons

- **Scaling / Unit Conversion:** If data is scaled: $\tilde{x} = c_1 x$ and $\tilde{y} = c_2 y$. $\tilde{\theta} = \frac{c_2}{c_1}\hat{\theta}$.
- **Swapping Covariate & Response:** If you mistakenly swap X and Y, the new slope $\hat{\theta}_{\text{swap}} \neq 1/\hat{\theta}$. By Cauchy-Schwarz: $\hat{\theta} \times \hat{\theta}_{\text{swap}} \leq 1$.
- **Duplicating Data:** If the entire dataset is duplicated (stacking X on X): $\tilde{X}^\top\tilde{X} = 2X^\top X$, $\tilde{X}^\top\tilde{Y} = 2X^\top Y$. The constant cancels out during inversion. **The optimal slope $\hat{\theta}$ does not change.**
- **Weighted Linear Regression:** Each observation has a predetermined weight $w_i \geq 0$. $\min_\theta \sum_{i=1}^{N} w_i(\theta\hat{x}_i - \hat{y}_i)^2$. **Derivative:** $\sum 2w_i(\theta\hat{x}_i - \hat{y}_i)\hat{x}_i$. **Optimal Slope (no intercept):** $\hat{\theta} = \frac{\sum w_i\hat{x}_i\hat{y}_i}{\sum w_i\hat{x}_i^2}$.
- **Robustness (OLS vs. LAD): OLS (Ordinary Least Squares):** Minimizes $(kx_i - y_i)^2$. Heavily influenced by **outliers** because errors are squared. **LAD (Least Absolute Deviations):** Minimizes $|kx_i - y_i|$. The optimal k is the **median** of the ratios $\{y_i/x_i\}$. It is highly **robust** to extreme outliers because it penalizes errors linearly.
- **Pipeline Rules:** Outliers $\implies$ use LAD. Too many features $\implies$ use LASSO.
- **Correlation vs. Regression: Correlation (r):** Measures the *strength* and *direction* of the linear relationship (strictly between -1 and 1). **Regression (θ):** Measures the *expected* change in Y for a 1-unit change in X ($\Delta y = \theta\Delta x$).

7. Fundamentals & Linear Models

Predictors & Engineering

- **The Linear Predictor:** $f(x) = \sum_{k=1}^{d} \hat{\phi}^{(k)}x^{(k)} = \hat{\theta}^\top x$. $x \in \mathbb{R}^d$: Covariate (input features). $\hat{\theta} \in \mathbb{R}^d$: Parameters (weights). **Ordinary Least Squares (OLS) Magic Formula:** $\hat{\theta} = (X^\top X)^{-1}X^\top Y$.
- **Mean Squared Error (MSE):** Measures fit to historical data (In-Sample Error). $\text{MSE} = \frac{1}{N}\sum_{i=1}^{N} \hat{e}_i^2$ where $\hat{e}_i = \hat{y}_i - \hat{\theta}^\top\hat{x}_i$.
- **Feature Engineering:** Using feature maps (Φ) to fit nonlinear curves using linear math. $\Phi(x_i) = \hat{x}_i = [\phi^{(1)}(x_i), \phi^{(2)}(x_i), \ldots, \phi^{(d)}(x_i)]^\top$. e.g., Polynomial features: $\phi^{(k)}(x) = x^k$. Creates high-dimensional feature vectors from 1D inputs.

8. Bias-Variance Trade-Off

Model Complexity

- Expected Prediction Error on unseen data: Error = Bias^2 + Variance + Noise Var.
- **Underfitting (High Bias):** Model too simple (e.g., degree 1 polynomial). Fails on both train and test data. **Fix:** Make model complex (more features), run optimization longer, decrease regularization.
- **Overfitting (High Variance):** Model too complex (e.g., degree 15 polynomial). Memorizes noise. Low train error, high test error. **Fix:** Make simpler (fewer features), get more data, stop optimization early, **Increase Regularization.**

9. Regularization & Norms

Shrinkage & Penalties

- **Occam's Razor:** Simpler models are preferred. We combat overfitting via **Shrinkage** (restricting the magnitude of parameters θ to ignore noise).
- **Mathematical Norms ($\|v\|_p$):** For a vector v of length d: **0-norm (L_0):** Counts non-zero elements. $\|v\|_0 \leq 1$ means max 1 non-zero item (shape: Cross/Axes). **1-norm (L_1):** Sum of absolute values. $\sum|v_i|$ (shape: Diamond). **2-norm (L_2):** Euclidean distance. $\sqrt{v^\top v}$ (shape: Circle). **∞-norm (L_∞):** Max absolute value. $\max|v_i|$ (shape: Square).
- **Ridge Regression (L_2 Regularization):** $\min_{\theta\in\mathbb{R}^d} \|X\theta - Y\|_2^2 + \alpha\|\theta\|_2^2$. α: Trade-off parameter. If $\alpha = 0 \implies$ OLS. If $\alpha \to \infty \implies \hat{\theta} \to 0$ (uninformative). **Rule:** Do NOT regularize the intercept (θ_0). It only shifts the baseline. **Closed-form solution:** $\hat{\theta}_{\text{ridge}} = (X^\top X + \alpha I)^{-1}X^\top Y$. **Invertibility:** $X^\top X + \alpha I$ is always invertible if $\alpha > 0$.
- **Prior Penalty (Macau Problem):** Shrink to prior v instead of 0. $\min\|X\theta - Y\|_2^2 + \lambda\|\theta - v\|_2^2 \implies \hat{\theta} = (X^\top X + \lambda I)^{-1}(X^\top Y + \lambda v)$.
- **LASSO Regression (L_1 Regularization):** $\min_{\theta\in\mathbb{R}^d} \|X\theta - Y\|_2^2 + \alpha\|\theta\|_1$. **Key property:** Induces **sparsity** (forces many $\theta^{(k)} = 0$). Acts as automatic feature selector. **Why?** The elliptical error contour hits the sharp corners of the L_1 diamond constraint first, where coordinates are zero.

10. Model Selection

Testing & Validation

- **Core Rule:** Training error is too optimistic. Must test on unseen data.
- **Cross-Validation (CV): K-Fold CV:** Split data into K non-overlapping folds. Train on $K-1$ folds, test on 1 fold. Cycle K times. Aggregate expected MSE and std dev. **Leave-One-Out CV (LOOCV):** $K = N$. Train on $N-1$, test on 1. Good for small data, but high computational burden. Compare models using the exact same data splits. Pick model with lowest Expected MSE. Standard practice: $K = 5$.
- **Data Leakage:** Happens when training and test sets are correlated (e.g., same patient in both sets, time-series overlaps). Results in falsely optimistic error estimates. Always split by subject/time properly.

11. HW02 Matrix Calculus & Derivations

Advanced Mechanics

- **Vectorization Tricks:** Sum of products: $\sum\hat{x}_i\hat{y}_i = X^\top Y$. Mean centered inner product: $\sum(\hat{x}_i - \bar{x})(\hat{y}_i - \bar{y}) = (X - \bar{x}\mathbf{e})^\top(Y - \bar{y}\mathbf{e})$. Regularized sum: $\sum\hat{x}_i\hat{y}_i + \alpha\sum\hat{x}_i^2 = X^\top Y + \alpha X^\top X$. Absolute loss sum: $\sum|\theta\hat{x}_i - \hat{y}_i| = \|X\theta - Y\|_1$.
- **Generalized Tikhonov Regularization:** Objective: $\min\ell(\theta) = \|X\theta - Y\|_2^2 + \|\Gamma\theta\|_2^2$. **1. Expand data term:** $\|X\theta - Y\|_2^2 = (X\theta - Y)^\top(X\theta - Y) = \theta^\top X^\top X\theta - 2X^\top Y\theta + Y^\top Y$. **2. Expand penalty:** $\|\Gamma\theta\|_2^2 = (\Gamma\theta)^\top(\Gamma\theta) = \theta^\top\Gamma^\top\Gamma\theta$. **3. Quadratic Form ($\theta^\top A\theta + b^\top\theta + c$):** $A = X^\top X + \Gamma^\top\Gamma$, $b = -2X^\top Y$, $c = Y^\top Y$. **4. Convexity (Hessian):** $\nabla_\theta^2\ell(\theta) = 2A = 2(X^\top X + \Gamma^\top\Gamma)$. Since $M^\top M$ is always Positive Semi-Definite (PSD), the sum is PSD. Yes, it's convex. **5. Gradient $\to 0$ for optimal $\hat{\theta}$:** $2(X^\top X + \Gamma^\top\Gamma)\theta - 2X^\top Y = 0 \implies \hat{\theta} = (X^\top X + \Gamma^\top\Gamma)^{-1}X^\top Y$. **6. Recovery:** If $\Gamma = \sqrt{\lambda}I$, then $\Gamma^\top\Gamma = \lambda I$. Substituting gives standard Ridge formula.
- **Reweighting (WLS):** Linear prediction: $y_{pred} = x_{new}^\top\hat{\theta}$. For Weighted Least Squares, $\hat{\theta}_{WLS} = (X^\top SX)^{-1}X^\top SY$. **Inverse Var:** If noise $\sim \mathcal{N}(0, \sigma^2)$, optimal weights are $w_i = 1/\sigma^2$. $y_{pred} = x_{new}^\top(X^\top SX)^{-1}X^\top SY$. By grouping Y_i, the weighting function w is: $w(X, S, x_{new}, i) = [x_{new}^\top(X^\top SX)^{-1}X^\top S]_i$.
- **Laplace Distribution $\to$ Absolute Error:** Noise mechanism: $\hat{\epsilon}_i \sim \text{Laplace}(0, 1)$. Likelihood given by $L = \frac{1}{2}\exp(-|y_i - \theta x_i|)$. **1. Full Likelihood:** $\prod_{i=1}^{N}\frac{1}{2}\exp(-|y_i - \theta x_i|)$. **2. Log-Likelihood:** $\log(\prod) \to \sum$. $\log(L) = -N\log(2) - \sum|y_i - \theta x_i|$. **3. Optimization Equivalence:** Maximizing a negative sum is mathematically identical to **minimizing** a positive sum. $\max(-\sum|y_i - \theta x_i|) \iff \min\sum|y_i - \theta x_i|$. **Conclusion:** Assuming Laplace noise justifies L_1 absolute loss (LASSO).
- **Chinchilla Scaling (Linearizing Non-Linearity):** Model: $\hat{y}_i = E + A\hat{x}_i^{-\alpha} + B\hat{d}_i^{-\beta} + \epsilon_i$. If α, β are fixed via Grid Search, treat $\hat{x}_i^{-\alpha}$ and $\hat{d}_i^{-\beta}$ as standard linear features! Matrix format ($Y = X\theta + \epsilon$): $X = \begin{bmatrix} 1 & \hat{x}_1^{-\alpha} & \hat{d}_1^{-\beta} \\ \vdots & \vdots & \vdots \\ 1 & \hat{x}_N^{-\alpha} & \hat{d}_N^{-\beta} \end{bmatrix}$, $\theta = \begin{bmatrix} E \\ A \\ B \end{bmatrix}$. Calculate predicted loss to compare budget constraints.

12. Logistic Regression Fundamentals

Classification Models

- Used for predicting categorical/discrete labels (e.g., $y \in \{0, 1\}$).
- **The Sigmoid (Logistic) Function:** Maps any real number to a probability between 0 and 1. $\sigma(z) = \frac{1}{1+e^{-z}}$. **Key Mathematical Properties (from HW):** **Derivative:** $\sigma'(z) = \sigma(z)(1 - \sigma(z))$. **Symmetry:** $\sigma(-z) = 1 - \sigma(z)$.
- **Hypothesis Formulation:** Model probability of the positive class ($Y = 1$) as: $P(Y = 1|X = x) = \sigma(\theta^\top x) = \frac{1}{1+e^{-\theta^\top x}}$.
- **Log-Odds (Logit):** Odds is the ratio of probability of success to failure. Odds $= \frac{P(Y=1|X)}{P(Y=0|X)} = \frac{\sigma(\theta^\top x)}{1-\sigma(\theta^\top x)}$. Taking the logarithm yields a linear equation: $\log(\text{Odds}) = \theta^\top x = \theta_0 + \theta_1 x_1 + \ldots$
- **Boundary:** At threshold $\tau = 0.5$, boundary is exactly where $\theta^\top x = 0$.
- **LLM Reward Model (Bradley-Terry):** Predict text A beats B via probability $\sigma(\theta^\top z_A - \theta^\top z_B)$. Treat $z_{new} = (z_A - z_B)$ as the new feature, then minimize standard Log-Loss.

13. Maximum Likelihood Estimation (MLE)

Objective Functions

- We find θ by maximizing the likelihood of the observed data (or minimizing the negative log-likelihood).
- **Standard Labels $y \in \{0, 1\}$: Log-Likelihood Objective:** $\max_\theta \sum_{i=1}^{N} [y_i\log(\hat{y}_i) + (1 - y_i)\log(1 - \hat{y}_i)]$.
- **Minimization Form (Cross-Entropy Loss):** $\min_\theta \sum_{i=1}^{N} \left[-y_i\theta^\top x_i + \log(1 + e^{\theta^\top x_i})\right]$.
- **Modified Labels $y \in \{-1, +1\}$ (from HW):** Because $\sigma(-z) = 1 - \sigma(z)$, we can write a unified likelihood for both classes: Likelihood$(\theta|x_i, y_i) = \sigma(y_i\theta^\top x_i)$. **Minimization Problem:** $\min_\theta \sum_{i=1}^{N} \log(1 + e^{-y_i\theta^\top x_i})$.
- **Gradient Descent:** Update rule to find original θ: $\theta_{(t+1)} \leftarrow \theta_{(t)} - \alpha\nabla\ell(\theta_{(t)})$ Where α is the step size (learning rate) and $\nabla\ell(\theta)$ is the gradient (first derivative).

14. Performance Metrics

Evaluating Classifiers

- Accuracy is unreliable for imbalanced data. Guessing majority gives high accuracy but is useless. We use the **Confusion Matrix** for better metrics.
- **Confusion Matrix Setup: TP (True Positive):** Actual 1, Predict 1. **FP (False Positive):** Actual 0, Predict 1. **FN (False Negative):** Actual 1, Predict 0. **TN (True Negative):** Actual 0, Predict 0. Total Actual Positives (P) = TP + FN. Total Actual Negatives (N) = FP + TN.
- **Key Metrics: Accuracy:** Overall correctness. Accuracy $= \frac{TP+TN}{P+N}$. **Precision:** When we predict positive, how often are we right? Precision $= \frac{TP}{TP+FP}$. **Recall (True Positive Rate/Sensitivity):** Out of all actual positives, how many did we find? Recall $= \frac{TP}{P} = \frac{TP}{TP+FN}$. **False Positive Rate (FPR):** Out of all actual negatives, how many false alarms? FPR $= \frac{FP}{N} = \frac{FP}{FP+TN}$. **F1-Score:** Harmonic mean of Precision and Recall. Fails to account for True Negatives (TN). F1 $= \frac{2}{\frac{1}{\text{Precision}} + \frac{1}{\text{Recall}}} = 2\times\frac{\text{Precision}\times\text{Recall}}{\text{Precision}+\text{Recall}}$.
- **ROC Curve and AUC: ROC Curve:** Plots **TPR (Y-axis)** vs. **FPR (X-axis)** across all classification thresholds τ. **Perfect Classifier:** Top left corner $(0, 1)$. **AUC (Area Under Curve):** Aggregate metric independent of threshold. AUC=1 is perfect, AUC=0.5 is random guessing.

15. Bayes' Classifier

Probabilistic Decision

- Classifies by finding the class c that maximizes posterior probability: $\mathcal{C}(x) = \arg\max_c P(Y = c|X = x)$. Using Bayes' Rule, and dropping the shared denominator $P(X)$: $\arg\max_c \left[P(X = x|Y = c)P(Y = c)\right]$. $P(Y = c) = \frac{N_c}{N}$ (Class proportion). $P(X|Y = c)$ is assumed to be a Gaussian Distribution.
- **Laplace Bayes:** Evaluates probability based on absolute distance $|x - \mu|$.
- **Poisson Bayes:** Likelihood $P(X = k) = \frac{\mu^k e^{-\mu}}{k!}$. Note: $e^{-\text{large}}$ dominates the calculation and effectively zeroes out probability.

16. Generative Models: LDA / QDA

Discriminant Analysis

- **Linear Discriminant Analysis (LDA): Assumption:** Both classes share the **same Covariance Matrix** ($\hat{\Sigma}_0 = \hat{\Sigma}_1 = \hat{\Sigma}$). Because covariances are equal, quadratic terms cancel out during log transformation. **LDA Decision Boundary (Matrix Math):** Predict Class 1 if: $2(\hat{\mu}_1 - \hat{\mu}_0)^\top\hat{\Sigma}^{-1}x \geq C$ Where the Constant C is: $C = 2\log\left(\frac{N_0}{N_1}\right) + \hat{\mu}_1^\top\hat{\Sigma}^{-1}\hat{\mu}_1 - \hat{\mu}_0^\top\hat{\Sigma}^{-1}\hat{\mu}_0$. *Note: Because x has no exponent, the boundary is a straight line/hyperplane in the feature space.*
- **Quadratic Discriminant Analysis (QDA): Assumption:** Classes have **different Covariance Matrices** ($\hat{\Sigma}_0 \neq \hat{\Sigma}_1$). Because covariances differ, the quadratic x^2 terms do NOT cancel out. **QDA Decision Boundary:** Predict Class 1 if: $x^\top(\hat{\Sigma}_0^{-1} - \hat{\Sigma}_1^{-1})x + 2(\hat{\Sigma}_1^{-1}\hat{\mu}_1 - \hat{\Sigma}_0^{-1}\hat{\mu}_0)^\top x \geq C$ Where Constant C involves determinants: $C = 2\log\left(\frac{N_0}{N_1}\frac{\sqrt{\det\hat{\Sigma}_0}}{\sqrt{\det\hat{\Sigma}_1}}\right) + \hat{\mu}_1^\top\hat{\Sigma}_1^{-1}\hat{\mu}_1 - \hat{\mu}_0^\top\hat{\Sigma}_0^{-1}\hat{\mu}_0$. *Note: Because of the $x^\top(\ldots)x$ term, the boundary is non-linear (parabolic/elliptical).*
- **Matrix Calculation Guide (from HW):** When given vectors and an identity matrix: **1. Means ($\hat{\mu}$):** Find the average of x_1 values and x_2 values for each class independently. **2. Inverse Covariance ($\hat{\Sigma}^{-1}$):** If $\hat{\Sigma} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, its inverse is also $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$. **3. Vector Transpose ($^\top$):** Turns a column vector $\begin{pmatrix} a \\ b \end{pmatrix}$ into a row vector $(a \quad b)$. **4. Dot Product:** $u^\top v = (u_1 \times v_1) + (u_2 \times v_2)$. E.g., $\hat{\mu}_1^\top\hat{\mu}_1 = (5 \times 5) + (4 \times 4) = 41$.

17. Support Vector Machines (SVM)

Hyperplane Basics & Geometry

- A hyperplane separates data into positive $(+1)$ and negative (-1) classes. Let x be the input vector, θ (or w) be the normal vector, and b be the bias. **Equation:** $\theta^\top x + b = 0$. **Prediction Rule:** $\hat{y}_{new} = \text{sign}(\theta^\top x_{new} + b)$. Predict $+1$ if $\theta^\top x_{new} + b \geq 0$. Predict -1 if $\theta^\top x_{new} + b < 0$.
- **1D Non-Separable Rule:** Sequence + - + needs 2 cuts. A 1D SVM can only make 1 cut, so max accuracy is 2/3 (sacrifices the minority point).

Margins & Distance Calculation

- **Functional Margin (γ_f):** Raw score of a point. $\gamma_f = y_i(\theta^\top x_i + b)$. **Geometric Margin ($\gamma$):** Actual perpendicular distance from the point to the boundary. $\gamma = \frac{\gamma_f}{\|\theta\|_2} = \frac{y_i(\theta^\top x_i + b)}{\|\theta\|_2}$. **Distance from point to line (2D):** For a line $Ax_1 + Bx_2 + C = 0$, distance from (x_1, x_2) is: Dist $= \frac{|Ax_1 + Bx_2 + C|}{\sqrt{A^2 + B^2}}$.
- **Total Margin Width:** Distance between the positive buffer ($\theta^\top x + b = 1$) and negative buffer ($\theta^\top x + b = -1$) is $\frac{2}{\|\theta\|_2}$.

Optimization & Formulation

- **Primal Optimization Problem:** Maximizing the margin $\frac{2}{\|\theta\|_2}$ is equivalent to minimizing $\|\theta\|_2^2$. **Hard-Margin SVM (Linearly Separable):** $\min_{\theta,b}\frac{1}{2}\|\theta\|_2^2$ s.t. $y_i(\theta^\top x_i + b) \geq 1\ \forall i$. **Soft-Margin SVM (Non-Separable):** Introduces slack variables $s_i \geq 0$ representing the penalty for misclassification. $\min_{\theta,b}\|\theta\|_2^2 + \lambda\sum_{i=1}^{N} s_i$ s.t. $y_i(\theta^\top x_i + b) \geq 1 - s_i\ \ \forall i$. *Note:* If $s_i = 0$, point is outside or on the margin. If $s_i > 0$, point has violated the margin buffer (but might not have crossed the center decision boundary yet). **Tuning λ:** High $\lambda \implies$ heavily penalizes errors (hard margin, overfit). Low $\lambda \implies$ forgiving, wider margin (robust).
- **Dual Formulation (The Math Trick):** Converts $d + 1$ variables into N variables (α). **1. Lagrangian Function ($\mathcal{L}$):** $\mathcal{L}(\theta, b, \alpha) = \|\theta\|_2^2 - \sum_{i=1}^{N}\alpha_i[y_i(\theta^\top x_i + b) - 1]$. **2. First Order Conditions (FOC):** Take derivatives with respect to θ and b, set to 0. $\nabla_\theta\mathcal{L} = 0 \implies \theta = \frac{1}{2}\sum\alpha_i\mathbf{y_i}\mathbf{x_i}$. $\nabla_b\mathcal{L} = 0 \implies \sum\alpha_i y_i = 0$. **3. Dual Optimization Problem:** Substitute FOC back into $\mathcal{L}$ to get: $\max_{\alpha\geq 0}\sum_{i=1}^{N}\alpha_i - \frac{1}{4}\sum_{i,j=1}^{N}\alpha_i\alpha_j y_i y_j x_i^\top x_j$ s.t. $\sum\alpha_i y_i = 0$.
- **Recovering the Classifier & Adjustments: Support Vectors:** Points where $\alpha_i > 0$. They lie exactly on the buffer boundaries. **Complementary Slackness:** To find bias b, pick any support vector ($\alpha_i > 0$). It satisfies: $y_i(\hat{\theta}^\top x_i + \hat{b}) - 1 = 0 \implies \hat{\mathbf{b}} = \mathbf{y_i} - \hat{\theta}^\top\mathbf{x_i}$. **Modifying Slope ($\Delta\theta$):** To fix a misclassified point (x_m, y_m), minimizing $\|\Delta\theta\|_2^2$ s.t. $\Delta\theta^\top x_m \geq c$ yields $\Delta\theta = \frac{\lambda}{2}x_m$. **Scaling Parameters:** If you multiply weights 2θ and $2b$, accuracy and boundary do *not* change. Number of physical support vectors does not change. **SVs Only:** Deleting non-support vectors does NOT change boundary. **Symmetric Points trick:** e.g., $(1, 0)^+, (0, 1)^- \implies \theta = (1, -1), b = 0$.

18. Decision Trees & ID3 Algorithm

Tree Concepts & Metrics

- **Tree Concepts: Root Node:** The top question. **Non-leaf Nodes:** Testing conditions (splits). **Leaf Nodes:** Final predictions.
- **Entropy (H):** Measures the "messiness" or uncertainty of a variable Y. $H(Y) = -\sum_{k=1}^{K} p_k\log_2(p_k)$. p_k: probability of class k occurring. **Convention:** $0\times\log_2(0) = 0$. If pure ($p = 1$ or 0), $H(Y) = 0$. If perfectly mixed ($p = 0.5$ in 2-class), $H(Y) = 1$.
- **Conditional Entropy ($H(Y|L)$):** Measures uncertainty of Y *after* knowing feature L. $H(Y|L) = \sum_l P(L = l) \times H(Y|L = l)$. **Steps to calculate: 1. Split** data into subsets based on feature L categories. **2. Calculate** $H(Y)$ for each subset individually. **3. Multiply** each subset's entropy by its probability (size of subset / total size), and sum them up.
- **Information Gain (IG):** Quantifies how much information L tells us about Y. $IG(Y, L) = H(Y) - H(Y|L)$. **ID3 Heuristic Rule:** At each node, calculate the Information Gain for all available features. **Choose the feature with the highest Information Gain to split.**

Stopping Criteria & Characteristics

- **Stopping Criteria (Edge Cases):** Stop recursing and convert the current node to a Leaf if: **1.** All samples at the node have the same label (pure). **2.** No remaining features to test (Predict majority label). **3.** No samples at this node (Predict majority label of the parent node).
- **Characteristics of Decision Trees: Advantages:** Highly intuitive / interpretable (like a flowchart). Fast and simple to implement (heuristic-based). No prior assumptions on data distribution. **Disadvantages:** Univariate splits: Only draws straight horizontal/vertical lines. Struggles to learn simple linearly separable data (e.g., diagonal lines). Pruning/tuning is tricky; prone to overfitting. Greedy algorithm: makes locally optimal choices; not guaranteed to find the globally perfect tree.

19. Bagging (Bootstrap Aggregating)

Averaging & Algorithms

- **Goal:** Reduce **variance** (overfitting) by building multiple independent models in parallel. Bias remains relatively unchanged.
- **The Math of Averaging:** If Z_m has mean μ and variance σ^2, the average $Y = \frac{1}{M}\sum Z_m$ has: **Mean:** $\mathbb{E}[Y] = \mu$ (Bias does not change). **Variance:** $Var(Y) = \frac{\sigma^2}{M}$ (Variance reduces by M).
- **Algorithm: 1. Bootstrapping:** Sample a dataset of size N *with replacement* to create M new datasets $\mathcal{D}_1, \ldots, \mathcal{D}_M$. **2. Train:** Train a model f_m on each dataset independently. **3. Aggregate:** For a new data point x_{new}: **Regression:** Average $\rightarrow \frac{1}{M}\sum f_m(x_{new})$. **Classification:** Majority Voting.
- **Random Forest:** An extension of bagging using Decision Trees. **Trick:** At each node, only consider a random subset of features. This reduces correlation between models, maximizing variance reduction.

20. Boosting

AdaBoost Algorithm (Binary)

- **Goal:** Reduce **bias** (underfitting) by building models **sequentially**. Each new model focuses on the mistakes of the previous one.
- **AdaBoost Algorithm: Uses Weak Learners** (e.g., Decision Stumps: a tree with only 1 split, high bias, low variance). Labels $y \in \{-1, +1\}$. **1. Initialize Weights:** $w_i = 1/N\ \ \forall i$. **2. For iteration $m = 1 \ldots M$:** a. Train weak learner $\mathcal{C}_m$ using current weights. b. **Weighted Error:** Sum weights of misclassified points: $err_m = \sum_{\mathcal{C}_m(\hat{x}_i)\neq\hat{y}_i} w_i$. c. **Model Weight:** Give good models higher say: $\alpha_m = \frac{1}{2}\ln\left(\frac{1-err_m}{err_m}\right)$. d. **Update Sample Weights:** $w_i \leftarrow w_i\exp(-\alpha_m\hat{y}_i\mathcal{C}_m(\hat{x}_i))$. *(If prediction is wrong, $\hat{y}_i\mathcal{C}_m = -1 \implies$ weight grows. If correct, weight shrinks.)* e. **Normalize:** Divide weights by $\sum w_i$ so they sum to 1. **3. Final Prediction:** $\mathcal{C}(x_{new}) = \text{sign}\left(\sum_{m=1}^{M}\alpha_m\mathcal{C}_m(x_{new})\right)$.

21. K-Means Clustering

Formulations & Heuristics

- **Goal:** Unsupervised learning. Group unlabeled data into k distinct, non-overlapping clusters based on similarity.
- **Standard K-Means Formulation:** Minimize the sum of squared Euclidean distances to cluster centers (c_j): $\min_{c_j, S_j}\sum_{j=1}^{k}\sum_{i\in S_j}||\hat{x}_i - c_j||_2^2$.
- **Alternating Optimization Heuristic: 1. Initialize:** Pick k random points as centers. **2. Fix Centers, Update Partitions:** Assign each point to the closest center. $S_j = \{i : ||\hat{x}_i - c_j||$ is minimized$\}$. **3. Fix Partitions, Update Centers:** Calculate the new mean of each group. $c_j = \frac{1}{|S_j|}\sum_{i\in S_j}\hat{x}_i$. **4.** Repeat Steps 2 & 3 until convergence. *Note: K-means guarantees convergence but may get stuck in a local optimum depending on random initialization.*
- **Weighted K-Means (Homework):** If samples have importance weights w_i: **Objective:** $\min\sum_{j=1}^{k}\sum_{i\in S_j} w_i||\hat{x}_i - c_j||_2^2$. **Center Update:** $c_j = \frac{\sum_{i\in S_j} w_i\hat{x}_i}{\sum_{i\in S_j} w_i}$.

22. Principal Component Analysis

Dimensionality Reduction

- **Goal:** Dimensionality reduction (d-dim to k-dim) for data compression/visualization while retaining maximum information.
- **Mathematical Equivalencies: Maximum Variance (Hotelling, 1933):** Maximize the spread of the projected points. $\max_{||v||=1} v^\top\Sigma_{\hat{X}}v$. **Minimum Error (Pearson, 1901):** Minimize the reconstruction error: $\min_{||v||=1}\mathbb{E}[||\hat{X} - \vec{Z}||_2^2]$. Both formulations yield the exact same solution.
- **PCA Algorithm Step-by-Step: 1. Standardize / De-mean:** Calculate mean vector: $\hat{\mu} = \frac{1}{N}\sum x_i'$. Subtract mean: $\hat{x}_i = x_i' - \hat{\mu}$. **2. Compute Covariance Matrix $\Sigma_{\hat{X}}$:** $\Sigma = \frac{1}{N}X^\top X$. **3. Eigenvalue Decomposition:** Solve for eigenvalues λ and eigenvectors v: $\det(\Sigma - \lambda I) = 0$ and $\Sigma v = \lambda v$. **4. Select Components:** Pick the top-k eigenvectors corresponding to the **largest** eigenvalues. Vectors must be **orthogonal** ($v_i^\top v_j = 0$) and **unit-length** ($||v||_2 = 1$). **5. Projection:** Project the original points onto the new k-dimensional subspace: $\hat{z}_i = (\hat{x}_i^\top v_1)v_1 + \cdots + (\hat{x}_i^\top v_k)v_k$. The scalar coordinate on vector v is just the dot product: $\hat{x}_i^\top v$.
- **Key Metrics & Properties: Lagrangian Formulation:** $\mathcal{L}(v, \beta) = v^\top\Sigma_{\hat{X}}v + \beta(1 - ||v||_2^2)$. **Rotation Equivariance:** If original data is rotated by α degrees, the principal components are also rotated by α degrees.